

Federated Capsule Graph Neural Network with Personalization

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ABSTRACT

Capsule-based GNN models have recently shown their competitiveness in graph classification. In this work, we propose FedCapsGNN, a novel framework which enables multiple clients to collaboratively perform capsule-based graph classification tasks under the federated learning paradigm. Our framework effectively combats the ill-effects of graph statistical heterogeneity with a novel local primary capsule augmentation method and a dexterously designed aggregation strategy with clustered capsule network layers. Experimental results demonstrate that our FedCapsGNN framework is able to capture the personality of each local client while effectively utilizing global knowledge, achieving superior accuracy for graph classification tasks on multiple benchmark datasets.

CCS CONCEPTS

• **Computing methodologies** → *Knowledge representation and reasoning*.

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1 INTRODUCTION

Graph classification is one of the most important problems and has been used in a wide variety of scientific fields, including bioinformatics, cheminformatics and AI medicine, etc [7, 19, 24, 25]. To solve this problem and inspired by CapsNet [11], capsules-based GNNs (CapsGNNs) [26, 30] have shown superiority in generating high-quality graph embeddings, which are able to capture the underlying meaningful information and the part-whole relationship of graph data. Different from the traditional GNNs [26] which perform

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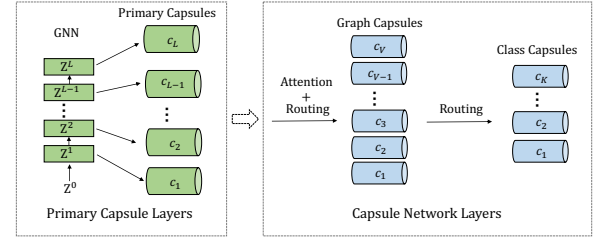


Figure 1: The architecture of the Capsule GNN.

aggregation of node features in the local neighborhood, the CapsGNNs are established on the routing mechanism to capture highly informative output in a small vector in place of a scalar output. Generally, CapsGNNs are comprised of the primary capsule layers and the capsule network layers as shown in Fig. 1. The primary capsule layers are based on GNNs and they are used to extract the primary capsules, which can be regarded as a special format of node features. The capsule network layers are derived from CapsNet and can be used to generate high-level graph or class capsules. These capsules reflect the properties of the graph from different aspects and may potentially capture more information of global properties, thus avoiding generating sub-optimal graph embeddings.

As in real-world applications, both GNNs and CapsGNNs need enough labeled graph data for the training process to obtain desired results. However, the acquisition of sufficient labeled graph data is time-consuming or even difficult, making the locally learned model far from satisfactory. For example, judging the efficacy of drugs based on graph representation requires adequate clinical research and trials. To solve this problem, we resort to Federated learning (FL), which is an emerging learning technique that allows distributed data owners (e.g., institutes and companies) to benefit from collaborative training while keeping their private data locally [18]. The main motivation of FL in graph learning is the shortage of training data for each single institute.

Recently, there have been some methods catching our sight which introduce the federated setting for GNNs [2] to solve the problem of poor model performance due to insufficient labeled graph data. So far, there hasn't been any research observing the situation where CapsNet or CapsGNNs are deployed in the federated setting yet. Despite that, directly combining CapsGNNs and traditional FL algorithms, e.g., FedAvg, is also confronted with limitations in coping with statistical data heterogeneity (also non-IID data) among clients as demonstrated in our empirical results. [29] provides a

detailed analysis of the the ill effects of the non-IID graphs in graph classification tasks and measures the levels of graph heterogeneity from the perspectives of graph structures and node features. However, it ignores the label distribution skew across local clients. For graph classification tasks, skewed label distribution of clients is more common in reality. For example, the label distribution in the text graph data of news media organizations and scientific journals tends to be different. Likewise, the distribution of patient categories will vary between oncology hospitals and polyclinics.

Therefore, it is of significant importance to develop client-specific models while exploiting the underlying shared information to tackle with the non-IID challenges mentioned above. To this end, we delicately design a personalized federated learning framework for capsule graph neural network (CapsGNN) in order to solve graph classification tasks with non-IID data distributions. Overall, we decouple the general CapsGNN into the global layers and the personalized layers. Specifically, corresponding to the CapsGNN mentioned above Fig. 1, it refers that we take the primary capsule layers as the global layers so that clients could benefit from global knowledge of all clients in primary capsule learning. And the capsule network layers will play the role of personalized part, which could further benefit from group-wise knowledge of the clients in the same group. Different from previous work [1, 4], we update the personalized layers by clustering clients into multiple groups and conducting group-wise averaging of the capsule network layers. This strategy is crucial as the locally learned capsule network layers are prone to over-fitting for clients with scarce data, and similar clients in the same group can benefit each other by sharing the personalized layers. Moreover, to further improve the generalization ability of personalized models, we also introduce a simple-yet-effective augmentation method by extracting primary capsules from the local primary capsule layers of the previous round, which are also referred to hereafter as historical primary capsule layers, to enhance the training of personalized layers. To the best of our knowledge, this is the first time that the capsule network is introduced into federated learning. The main contributions of this work can be summarized as follows:

- We propose a novel personalized FL framework for Capsule GNN learning with global primary capsule layers and clustered capsule network layers to boost graph classification performance in non-IID settings.
- We present a novel local primary capsule augmentation method in the FL framework to enhance the local training of capsule network layers.
- We evaluate our framework on popular datasets and the results demonstrate the effectiveness and superiority of our framework.

2 RELATED WORK

The increasing concerns with data privacy have led to a rapid development of training GNNs in a federated setting [12, 20, 28]. In particular, [3] proposed a vertical federated GNN framework for privacy-preserving node classification and [10] proposed the SpreadGNN to extend federated multi-task learning to serverless settings for GNNs. To handle the non-IID graphs, clustered FL framework [29] and model-agnostic meta-learning [27] were also employed. Moreover, [9] implemented an open benchmark system

and [16] provided an overview for federated GNNs. While most existing works consider node classification and link prediction on graphs, our study focuses on graph classification tasks.

On the other hand, many personalized FL approaches have been developed in recent years, including multi-task learning [5, 15], meta-learning [6] and sharing only representation layers [1, 4]. In particular, clustered FL [8, 21, 23] is to group similar clients and train a individual global model for each group. Different from existing works, our method aims at graph learning tasks and exploits both shared representation layers and clustered classification layer to maximize the utility of shared knowledge across clients.

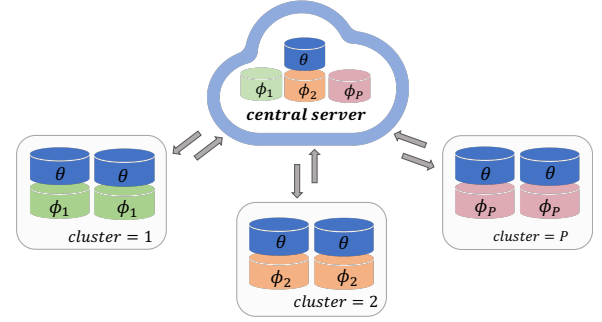


Figure 2: Global Aggregation with Clustered Capsule Network Layers

3 THE PROPOSED FRAMEWORK

3.1 Problem Formulation

In this section, we introduce our proposed framework as shown in Fig. 2. We assume there is one central server and m local clients and the server cluster the clients into a set of clusters $\{C_1, C_2, \dots, C_P\}$, $p \in [P]$ and P is the number of clusters. Each local client i owns a set of graphs G_i , where each graph $G_j \in G_i$ concludes N_j nodes, node features Z_j^0 and label y_j of the graph. The goal of graph classification is to learn a mapping $\mathcal{M} : \mathbb{G} \rightarrow \mathbb{Y}$, where $\mathbb{G}$ and $\mathbb{Y}$ are the spaces of graph data and label, respectively. We formulate the problem into the following optimization task:

$$\min_{(\mathbf{w}_1, \dots, \mathbf{w}_m)} \left\{ F(\mathbf{w}_1, \dots, \mathbf{w}_m) := \frac{1}{m} \sum_{i=1}^m F_i(\mathbf{w}_i) \right\}, \quad (1)$$

$$F_i(\mathbf{w}_i) = \frac{1}{|G_i|} \sum_{(G, y) \in G_i} \ell(\mathbf{w}_i; G, y), \quad (2)$$

where $\mathbf{w}_i$ denotes the personalized model for the i -th client and $\ell(\cdot; \cdot)$ is the loss function. Notice that for capsule-based GNNs, the model $\mathbf{w}$ can be decoupled into primary capsule extraction $z = f(G)$ parameterized by θ and classification decision $g(z)$ parameterized by ϕ . In our framework, we aim at learning a shared θ and a personalized ϕ for each client.

3.2 Capsule Graph Neural Network

Primary Capsule Layers. The primary capsule layers are used to generate primary capsules for nodes of a graph [30]. In our work, we choose the widely used GNN architecture proposed by [14] to

extract local features with different receptive-field of the graph nodes. This procedure can be written as:

$$\mathbf{Z}_j^{l+1} = \sigma(\sum_i \mathbf{T} \mathbf{Z}_i^l \mathbf{W}_{ij}^l), \quad \mathbf{T} = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}, \quad (3)$$

where $\mathbf{Z}_j^l \in \mathbb{R}^{N \times d}$ represents nodes features from the j th channel at the layer l , d is the number of feature channels and N is the number of nodes. $\mathbf{T} \in \mathbb{R}^{N \times N}$ is the information transform matrix and it is usually calculated from the adjacency matrix $\mathbf{A}$ for guiding the information flowing between nodes. $\mathbf{Z}_j^{l+1} \in \mathbb{R}^{N \times d'}$, $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\tilde{\mathbf{D}} = \sum_j \tilde{\mathbf{A}}_{ij}$. $\mathbf{W}_{ij}^l \in \mathbb{R}^{d \times d'}$ is the trainable matrix and serves as the channel filters from i th channel at the i th layer to the j th channel at the $(l+1)$ th layer. $\sigma(\cdot)$ is the activation function, e.g., $\tanh(\cdot)$.

Capsule Network Layers. The capsule network layers are used for classification decision, in which the attention module and dynamic routing mechanism are applied to assign primary capsules to obtain high-level graph capsules. Then applying dynamic routing mechanism again on the graph capsules can further get the class capsules, by which the classification decision can be made.

For the capsule network layers, both of the input u_i and the output c_j are vectors. We apply a transformation matrix W_{ij} to the capsule output u_i of the previous layer and transform u_i to $\hat{u}_{j|i}$ from V -dimension to K -dimension. Then we compute a weighted sum s_j with weights v_{ij} .

$$\hat{u}_{j|i} = W_{ij} u_i, s_j = \sum_i v_{ij} \hat{u}_{j|i}, \quad (4)$$

where v_{ij} are coupling coefficients that measure how likely capsule i may activate capsule j and then we apply a squashing function to s_j so that the final output vector c_j of the capsule has length between 0 and 1.

$$c_j = \frac{(\|s_j\|)^2}{1 + (\|s_j\|)^2} \frac{s_j}{\|s_j\|}. \quad (5)$$

3.3 Local Training with Primary Capsule Augmentation

Loss Function. Here followed by [22], we define the loss function $\ell(\cdot)$ as the sum of the classification loss $Loss_c$ that is calculated by margin loss and the reconstruction loss $Loss_r$ that is used as regularization. More specifically, they can be computed as follows:

$$Loss_c = \sum_{k=1}^K T_k \max(0, \mu^+ - \|c_k\|)^2 + \lambda(1 - T_k) \max(0, \|c_k - \mu^-\|)^2, \quad (6)$$

where $\mu^+ = 0.9$, $\mu^- = 0.1$ and $T_k = 1$ if the input graph belongs to class k . λ is used to stop initial learning from reducing the length of all class capsules.

$$Loss_r = \frac{\sum_i MP_i (d_i - \mu_i)^2}{\sum_i MP_i} + \frac{\sum_i (1 - MP_i) (d_i - \mu_i)^2}{\sum_i (1 - MP_i)}, \quad (7)$$

where μ_i represents the number of nodes with the attribute i appear in the input graph, d_i is the corresponding decoded value. $MP_i = 1$ if input graph contains nodes with attribute i .

Primary Capsule Augmentation. During the local training process, we conduct local primary capsules augmentation which are generated by the historical local primary capsule layers to enhance the training of personalized layers. Specifically, the local data are fed into the historical primary capsule layers, and the output primary capsules will also be utilized for training the capsule network layers simultaneously. The augmented primary capsules are

$$z' = f(\theta_i^{(t-1)}; G), \quad \forall G \in G_i. \quad (8)$$

where $\theta_i^{(t-1)}$ is the parameters of the i th historical primary capsule layers which are fixed in the current training round t . Then the final local optimization objective function F_i can be expressed by:

$$F_i(\mathbf{w}_i) = \frac{1}{|G_i|} \sum_{(G,y) \in G_i} \{ \ell[g(f(G)); y] + \alpha \cdot \ell[g(z'); y] \}, \quad (9)$$

where α is an adjustable hyper-parameter.

3.4 Global Aggregation with Clustered Capsule Network Layers

Client Clustering. The purpose of our clustering is to learn a better personalized model for each client through mutual promotion between similar clients. In our work, we evaluate the similarity of data distribution of clients by the capsules generated by each local model from a public dataset. Our clustering approach tends to be more stable and accurate compared to the previous methods which perform clustering by the gradients or parameters of models. And in realistic applications, it's possible to collect some unlabeled graph data as a publicly accessible dataset after desensitization. Therefore, we can measure the similarities between clients by computing the distance of the capsules generated by each local model, so as to achieve the clustering of clients. Here, we directly use the off-the-shelf clustering approaches, e.g., KMeans [17].

Decoupled Model Parameter Aggregation. Our method conduct different aggregation strategies for the common layers and the personalized layers as shown in Fig.2. More specifically, we use the FedAvg algorithm to average the local parameters to obtain the global shared primary capsule layers:

$$\theta \leftarrow \frac{1}{m} \sum_{i=1}^m \theta_i \quad (10)$$

and the parameters of capsule network layers will be averaged within each cluster as:

$$\phi_p \leftarrow \frac{1}{|C_p|} \sum_{j \in C_p} \phi_j, \quad p \in [P], \quad (11)$$

Algorithm 1 FedCapsGNN

Input: Number of clients m , number of clusters P , Communication rounds T , Local training epochs E , Step-size η .

Initialize: Initialize $\mathbf{w}^{(0)} = \{\theta^{(0)}, \phi^{(0)}\}$.

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1: for  $t = 0$  to  $T - 1$  do
2:   Server broadcasts  $\theta^{(t)}$  to each client  $i$  and  $\phi_p^{(t)}$  to clients in
   cluster  $C_p$ .
3:   for each client  $i$  in cluster  $C_p$  in parallel do
4:     Set  $\mathbf{w}_i^{(t)} = \{\theta^{(t)}, \phi_p^{(t)}\}$ 
5:     Local SGD  $\mathbf{w}_i^{(t+1)} = \mathbf{w}_i^{(t)} - \eta \nabla F_i(\mathbf{w}_i^{(t)})$  for  $E$  epochs
6:     Send  $\mathbf{w}_i^{(t+1)} = \{\theta_i^{(t+1)}, \phi_i^{(t+1)}\}$  to the server
7:   end for
8:   Server executes:
9:     Group the clients into  $P$  clusters.
10:    Update  $\theta^{(t+1)} \leftarrow \frac{1}{m} \sum_{i=1}^m \theta_i^{(t+1)}$ 
11:    Update  $\phi_p^{(t+1)} \leftarrow \frac{1}{|C_p|} \sum_{j \in C_p} \phi_j^{(t+1)}$ 
12: end for
    
```

3.5 Algorithm Design

Algorithm 1 presents the implementation process of our method. For the federated optimization process, to make each client obtains a high-quality model suitable for the local objective function, we divide the model update process into two parts: 1) joint update of primary capsule layers and 2) clustering update of capsule network layers. At each round, the clients will perform the local training in parallel and then send the locally updated parameters to the server for aggregation, while the server will group the clients into P clusters and broadcasts $\{\theta, \phi_p\}$ to clients in each cluster C_p .

4 EXPERIMENTS

In this section, we conduct experiments on various benchmark datasets to verify the improved performance of our proposed method over existing baseline methods on graph classification tasks.

4.1 Experimental Setup

In this study, we choose the following commonly used benchmarks, including biological and social graph datasets: D&D, COLLAB, IMDB-Binary (IMDB-B), IMDB-Multi (IMDB-M), Reddit-Multi (Reddit-M). It is worth noting that the social graphs have no node attributes, while the biological graphs have categorical node attributes. More details about the data statistics and properties can be found in [31].

In our experiments, we adopt 4 GCN layers to extract node features for the primary capsule layer and the number of output capsules in the capsule network layer is equal to the number of classes. α is set to 1 and the number of iterations in the dynamic routing is set to 3. We use a mini-batch size of 32, and employ Adam [13] as the local optimizer with initial learning rate 0.0001. For all FL methods, the numbers of global rounds T and local epochs E are set to 200 and 2; the number of clients M is set to 10. Similar to [21], we introduce the random partition to simulate the non-IID

scenario. Clients from the same group will be randomly assigned an array of class distribution $[p_1, p_2, \dots, p_K]$, where K is the number of classes and $\sum_{i=1}^K p_i = 1$. For example, a client may be randomly assigned with the partition of 0.1 : 0.2 : 0.7 in the three-category dataset. We adopt KMeans clustering method and the number of clusters is $P = M/2$.

4.2 Baseline Methods

To show the improvement of our proposed federated learning method, we compare our method with six baselines. (1) FedAvg [18]: the most widely used method in FL. (2) FedAvg-FT: a variant of FedAvg with local fine-tuning. (3) FedPer [1]: a model decoupling method that only aggregates the representation layers. (4) Ditto [15]: the state-of-the-art multi-task learning based method for personalized FL. (5) CFL [23]: a clustered FL method that learns multiple global models. (6) Standalone: clients train their models on local data without FL.

Table 1: Test accuracy (%) on benchmark datasets.

Method	COLLAB	IMDB-B	IMDB-M	D&D	Reddit-M
Standalone	66.41	78.89	54.56	54.36	49.60
FedAvg	65.45	73.33	61.75	64.76	50.65
FedAvg-FT	73.44	91.11	61.81	58.68	49.61
FedPer	67.26	81.11	61.15	73.20	48.20
Ditto	68.04	92.22	63.35	64.39	51.08
CFL	64.63	71.78	55.82	58.21	47.12
Ours	79.69	95.56	70.11	82.00	53.21

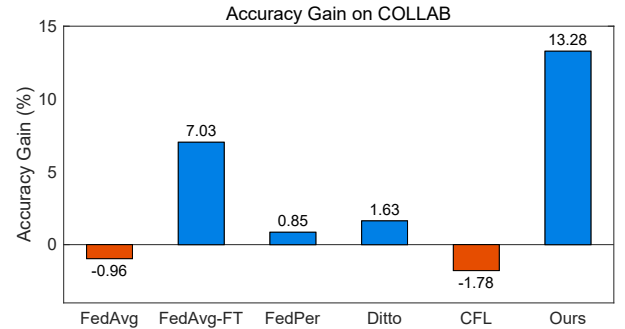


Figure 3: Accuracy gains compared to standalone on COLLAB.

4.3 Experimental Results

Classification Results: The average test accuracy results on the benchmark datasets are reported in Table 1 and Figure 3 shows the performance gains (over standalone) achieved by various FL methods on COLLAB. For each dataset, we highlight the top 2 results in bold. More specifically, we have the following observations from the results: (1) Our method can significantly improve the average accuracy of graph classification tasks and consistently achieve the best performance on various datasets. (2) The results demonstrate

that, in the non-IID scenario, some existing federated methods such as FedAvg may actually degrade the accuracy of the model.

Effect of System Size: To study the effect of different system sizes, we also test the baselines and our method on COLLAB dataset with 5, 10 and 20 clients, respectively. The results in Table 2 show that our method consistently outperforms all baselines. We also notice that as the number of clients increases, the accuracy results will decrease slightly. This is not surprising as larger number of clients means fewer data samples will be allocated to each client, which could result in enlarged distribution discrepancy among local data and a higher probability of local model over-fitting.

Table 2: Accuracy (%) under varying system sizes on COLLAB.

# Clients	FedAvg	FedAvg-FT	FedPer	Ditto	CFL	Ours
5 clients	76.25	76.54	76.07	76.66	77.12	83.41
10 clients	65.45	73.44	67.26	74.04	64.63	79.69
20 clients	54.76	62.79	65.90	73.45	63.92.22	79.12

Ablation Study: To better demonstrate our contributions, we further design an ablation study to verify the utility of different components of our method. We test our method without the primary capsule augmentation (CA) or client clustering (CC), respectively. The numerical results are reported in Table 3, from which we can see that the client clustering and the data augmentation in our method both play a significant role in improving the accuracy.

Table 3: Ablation experiments on COLLAB.

Method	COLLAB	IMDB-B	IMDB-M	D&D	Reddit-M
Ours	79.69	95.56	70.11	82.00	53.21
w/o CA	75.84	94.86	66.07	77.73	52.38
w/o CC	68.35	93.33	62.76	68.89	52.38

5 CONCLUSION

In this paper, we propose a novel personalized FL framework with CapsGNN model to provide a solution for graph classification tasks over distributed and heterogeneous data. We decouple the CapsGNN into the common primary capsule layers and personalized capsule network layers, and capture the relations between clients to enhance the learning of personalized layers. Experimental results on benchmarks demonstrate the effectiveness of our proposed framework. Future work could include exploring more real-world graph-based tasks in the non-IID scenarios with our framework.

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